



Mamaroneck Robotics 8490

Engineering Portfolio

2024-2025

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Meet The Team



Team Members:

Mauricio Villarreal, Danny McCance, Liam DeLaMora, Jordan Metsch, Alex Lin, Miles Contopoulos, Jonathan Wittick, Mustafa Khan, Nate Metsch, Maya Okochu, Hugh Scobie, Mika Torres, Luke Minchilli, Chloe Glazer, Max Lazarus

About Us:

Team 8490 represents Mamaroneck High School, in Mamaroneck, New York. The team started competing in the FIRST Tech Challenge in 2015 and since then, our school has implemented an impressive engineering program, and our team has grown with it.

The team meets several times a week where we design, program, and troubleshoot.

To better organize ourselves and improve efficiency, we split the team into three branches, though all members are welcome to move and help with whatever is needed. Our Engineering branch is responsible for designing, modeling, and building the robot, while Coding focuses on the software aspects. Documentation and Outreach handles community engagement and records.

While we have had our challenges through the years, our team has always persevered, and we look forward to challenging ourselves and growing our skills in the future.

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Team Goals

2024-2025 Goals

While our overarching goal has always been to do our best in the competitions and learn as a team, we have reevaluated how to best do that over time.

For this year, we wanted to focus on designing a simpler robot with better consistency, instead of flashier or specific attachments, which have led to issues both in the build process and the competitions. We wanted to make our robot as efficient as possible, utilizing designs like our claw and a revamped chassis.

In addition, our team hopes that our continued participation in the FTC competitions will grow our skills and knowledge as engineers and programmers. We have been looking to make a positive impact on our community, grow more united as a team, and have fun throughout the process.

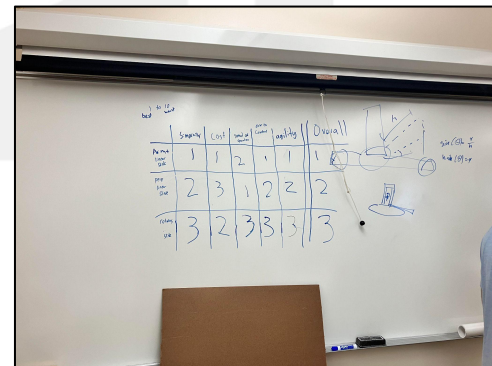
Design Process

Our design process always begins with lateral thinking. As a team, we brainstorm numerous ideas and then we decide on one that is efficient and plausible given the constraints. As one example, in the above photo, we ranked our top intake designs on simplicity, cost, speed, and agility.

Our claw and intake design required a lot of our energy towards the beginning of the season, and it was a significant challenge to decide on one, since this year's blocks have such a wide range of possible orientations. After evaluating our designs, which featured spinning wheels or rotating claws, we decided on the parallel claw.

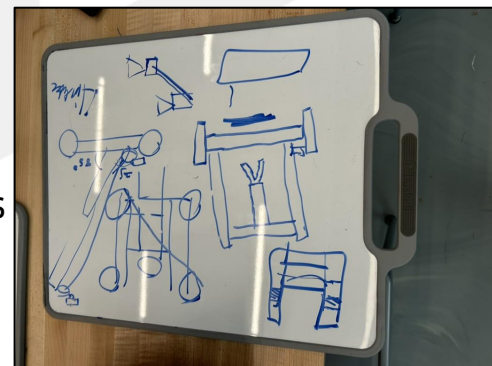
Another challenge was how we could attach the servo on the end of the wrist motor. Because this motor is set back from the actual pivot point, there was no good way to put the servo on the end. Eventually, we realized we could use a belt with a tensioner, which would allow us to fix it if it stretches out.

Despite the challenges, each new iteration is valuable in constructing our robot, and we make sure to enjoy the entire process.



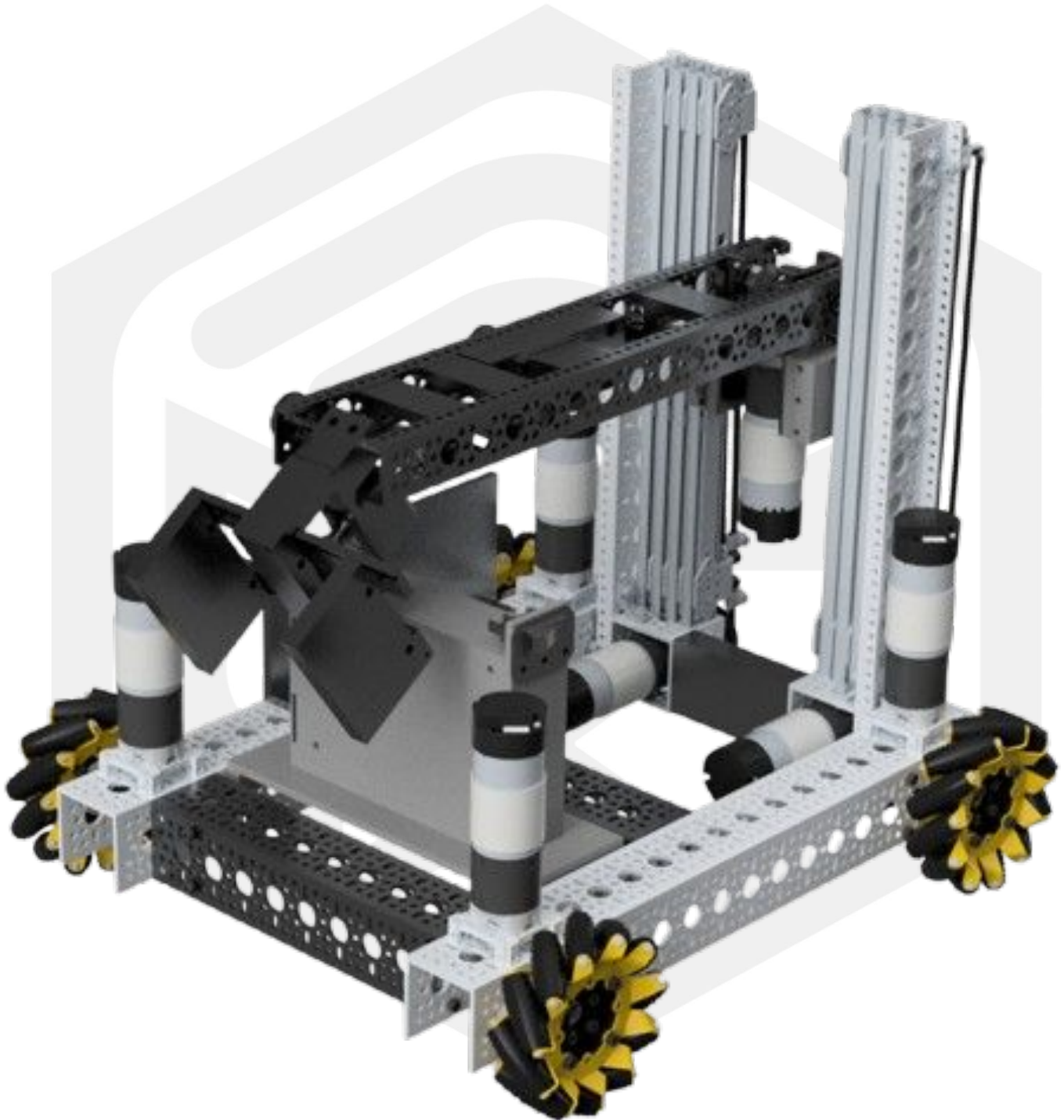
	Simplicity	Cost	Speed	Agility	Overall
Design 1	1	2	1	1	1
Design 2	2	3	1	2	2
Design 3	3	2	3	3	3

The team's rankings of different intake designs



One of our early ideas for the robot

Final Design



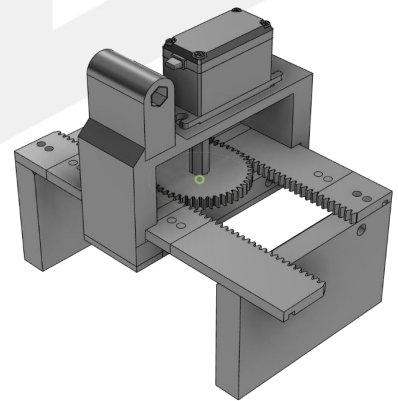
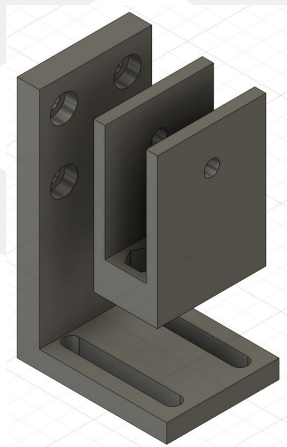
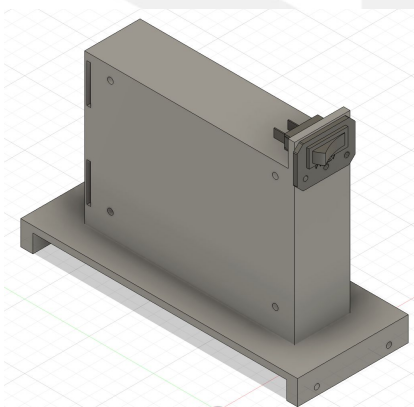
Custom Parts

Why custom parts?

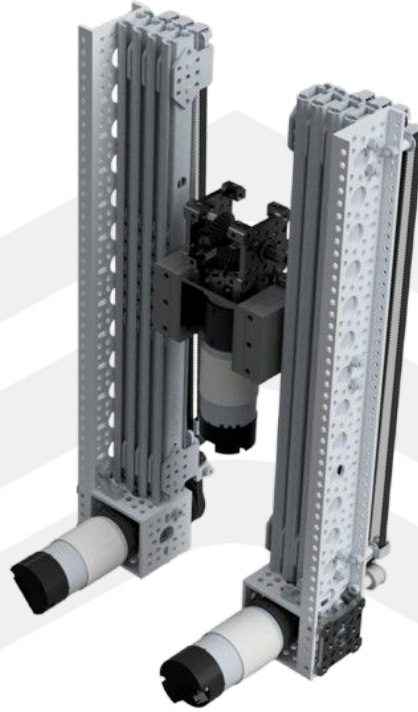
As a school club, we need to make do with what we have available to us, as we have limited funds and time.

Off-the-shelf parts do not meet the specific needs our robot has to succeed in the season's challenges. We heavily use 3D-printed parts designed with Fusion360 to both optimize our robots layout and compete in competition tasks. Our most notable 3D printed parts are a combined battery and control hub mount, spacers for linear slides, a belt tensioner, and our claw. Another 3D printed part are our custom side panels which attach to the robot using magnets. We also needed to build our own cable for the arm because we could not find a premade one of the right length that was coiled the way we needed. To do this we ran smaller wires through heatshrink and coiled it by heating it around a pipe. These parts both help us be more adaptable throughout our robot's design process and make us stand out.

CAD Models



Linear Slides



Design Process

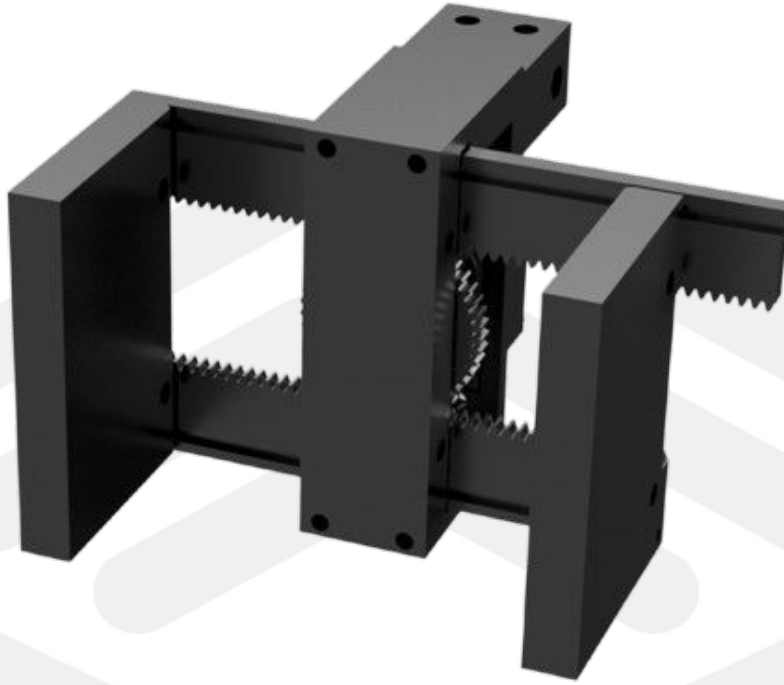
Our design process begins with lateral thinking. As a team, we brainstorm numerous ideas and narrow down till we determine the most efficient and feasible design. This initial phase was followed by extensive team discussions and brainstorming sessions, where various design ideas were evaluated based on factors such as complexity, efficiency, and feasibility within the given time frame. Through comparative analysis, including assessments of different types of robot arms like Claws, and intake that grab the block with vacuum like wheels, the team concluded that simplicity and efficiency were paramount. Ultimately, the decision to pursue a robotic arm with a claw at the end of it design over alternatives. We used a combination of old and new linear slides on our robot.'

Specifications

- 13.7:1 ratio gear 5203 Series GoBilda Yellowjacket motor
 - Rotates 150°
- Intakes and outtakes on opposite sides to save time

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Claw



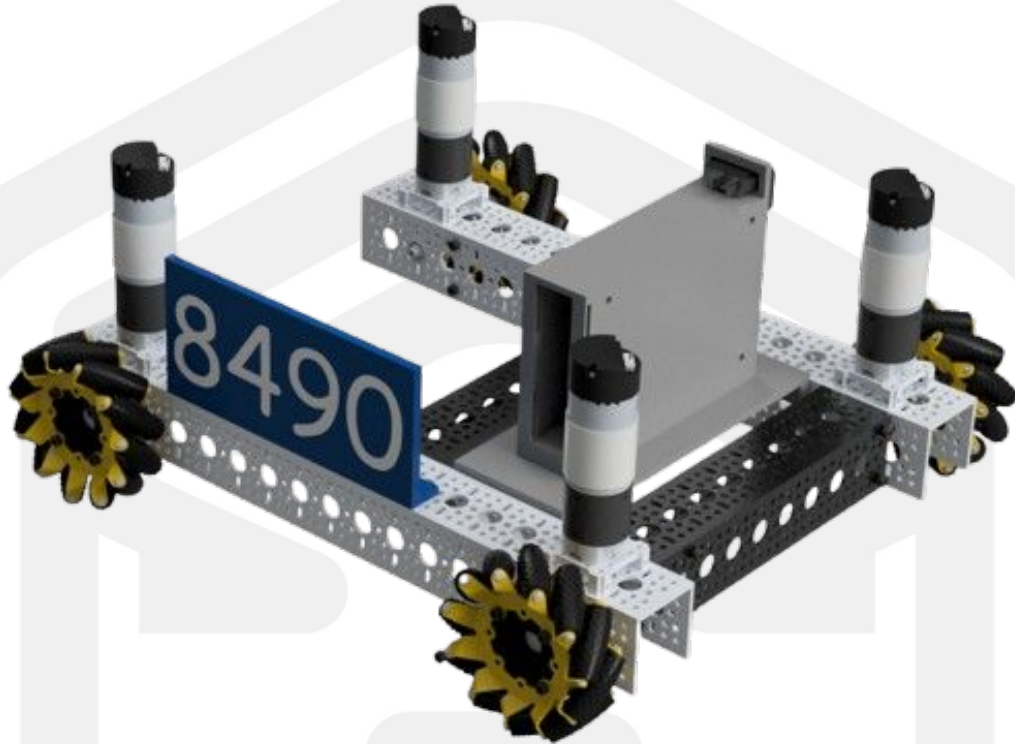
Design Process

The design process for our robotics team's bar lift involved several stages of brainstorming, prototyping, and iterative refinement. Initially, we considered a simple hook mechanism for lifting the bar, but realized that it wouldn't provide the necessary strength to lift the robot effectively. We then explored alternative options, eventually settling on a design centered around a lead screw driving a platform. This platform would be controllable in height and capable of hoisting the robot. However, we encountered issues with this design as it was too slow and required parts that were not readily available to us. Faced with constraints on ordering new parts, we pivoted to leveraging what we had on hand, including recycling parts from a robot design used two years prior. We also employed 3D printing to create winching mechanisms and other necessary components. The core of our final design was built around a linear slide with a c-channel attached to it. We carefully calculated the tension to ensure that the bar lift could handle the weight of the robot without compromising its structural integrity. One of the major challenges we encountered was mounting the bar lift onto the robot, which required careful consideration of motor configuration and finding the strongest point without affecting the overall performance. Once assembled, we focused on balancing the robot and conducting testing to determine the optimal placement for the bar lift to hang the robot effectively. Through this process of design, problem-solving, and resourcefulness, we were able to create a bar lift mechanism that met the requirements of our robotics competition while making the most of the resources available to us.

Specifications

- 13.7:1 ratio gear 5203 Series GoBilda Yellowjacket motor

Chassis



Design Process

As with previous years, we built the chassis primarily with GoBilda U-Channel. This year, we wanted to maximize space in the chassis for the linear slides and arm. To accomplish this, we decided to mount the control hub and batteries together with a vertical mount. To maximize space in the U-Channel for cable management we also mounted the motors for the wheels vertically using gears to connect them.

Specifications

- Chassis dimensions 45.5cm x 40.6cm
 - 96mm GoBilda mecanum wheels
- 5203 planetary gear series GoBilda yellow jacket motors with a 19.2:1 gear ratio and 312 rpm

Outreach

Outreach allows our team to give back to our community, make connections with local organizations, and inspire the next generation to pursue their interests in STEM.



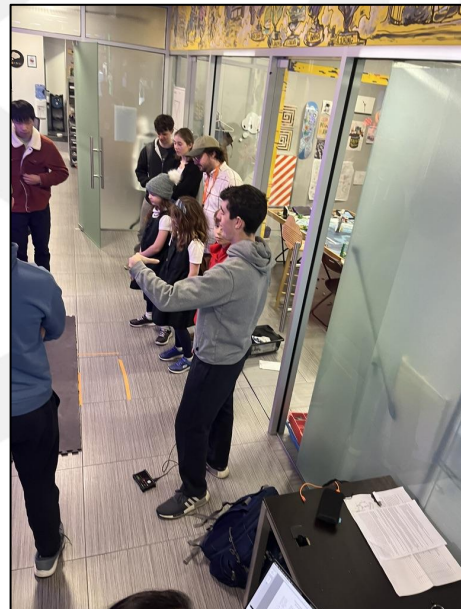
Alex guides the kids through running the robot



MakeInspires 2024

On January 7th, our FTC team organized an event with a local FLL team. Based at MakeInspires' Mamaroneck location, many of the FLL members will be entering the high school in a few years.

We shared our robot and process, as well as listening to their team's presentation and giving them advice.



Max explains the robot to the FLL team

Programming

Goals

When the programming team began coding the robot, they had two goals in mind: ease of operation and versatility. The most important thing for the success of our team was that the robot had to be easy to operate for any driver. From the start, we had ambitious goals to meet every task in the challenge, and the coding team made sure that every need was met. To strengthen the ease of operation, the programming team included many driver enhancements such as a toggle button for the intake, intuitive controls, and automatic positioning for the outtake.

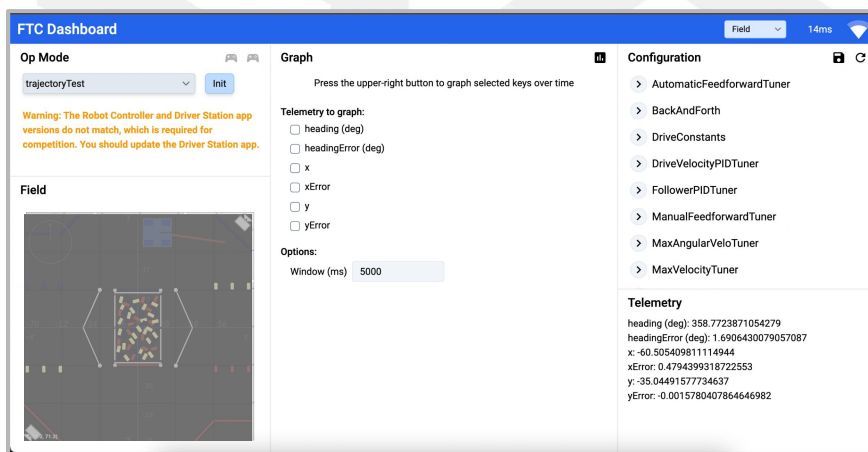


Miles works on the computer vision programming

How It Was Done

For both the TeleOp and autonomous portions, the robot was coded in Java using Android Studio. The autonomous portion utilized roadrunner in order to make easy and accurate paths for the robot to

follow during the autonomous phase of the competition. To program computer vision for the challenge, TensorFlow and machine learning models were used.



Roadrunner was used for the autonomous phase

TeleOp

```
while (opModeIsActive()){  
  
    telemetry.addData( caption: "X", -gamepad1.left_stick_y);  
    telemetry.addData( caption: "Y", gamepad1.left_stick_x);  
    telemetry.update();  
  
    move(gamepad1.left_stick_x, -gamepad1.left_stick_y, gamepad1.right_stick_x, speed);  
  
    if(gamepad1.right_bumper){ // intake motor control  
        intake.setPower(1);  
    }else{  
        intake.setPower(0);  
    }  
  
    if(gamepad1.left_bumper){ // spinner motor control  
        spinner.setPower(.65);  
    }else{  
        spinner.setPower(0);  
    }  
  
    if(gamepad2.dpad_down){  
        OuttakeLowerPos(telemetry, gamepad1);  
    }  
  
    if(gamepad2.dpad_right){  
        OuttakeMiddlePos(telemetry, gamepad1);  
    }  
  
    if(gamepad2.dpad_up){  
        OuttakeUpperPos(telemetry, gamepad1);  
    }  
}
```

The main TeleOp Code

Code

The TeleOp code was designed with scalability and readability in mind. The code has several custom classes such as the drivetrain class which allows all drive motors to be controlled by one line. The code also has several robust methods, including the one that controls our arm. In order to effectively control both linear slides, we had to use a new paradigm designed for dual-axis control.

Intuitive Driving

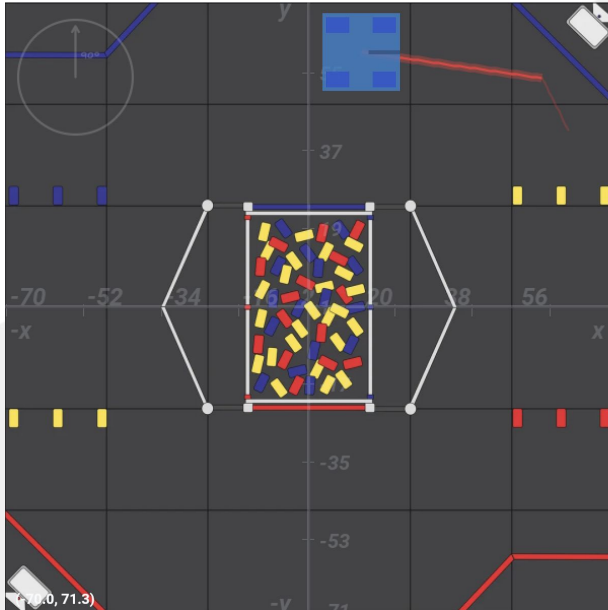
The team wanted to take as much work away from the driver and make driving as intuitive as possible. With this in mind we programmed semi autonomous functions into the TeleOp. This included pre-programmed positions for intaking samples and placing samples in each bucket and specimens on the high bar. This ensured that, during the TeleOp phase, the driver has to spend little time thinking about the linear slide, and can instead focus on ensuring safe robot operation and game strategy. The end result is a robust system that can consistently score in the high bucket, using any block on the field.

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Autonomous

Movement

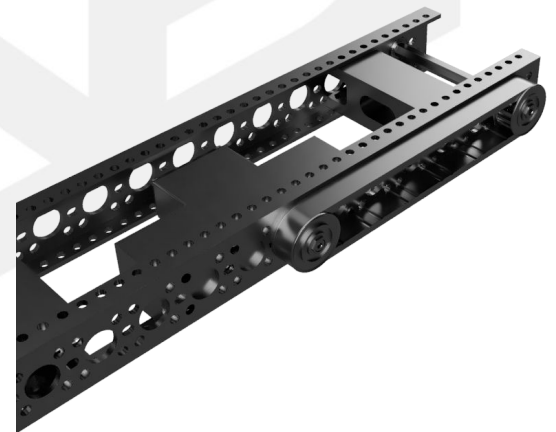
The team decided on using the Road Runner library to program autonomous movements because of its convenience and ability to program complex paths. Using data from PID loops, motor encoders and an IMU the robot is able to follow the paths very precisely. This year, the autonomous code places three blocks in the high bucket, or two specimens on the high bar, depending on alliance preferences.



Meep Meep is a program that visualizes Road Runner paths

Digital Parallel Bar

When mounting the servo on the arm, our team had two choices - a mechanically linked parallel four bar, or a digitally (servo) controlled four bar. Although a mechanical four bar would be simpler, the digital four bar gave us more control and flexibility over the rotation of the claw. To create the desired parallel-behavior, we programmed the wrist-servo to rotate in sync with the arm-motor, for part of the degree range. For the other half of the arm's range, the wrist autonomously maintains an appropriate position for scoring, which would be impossible without a digitally controlled mechanism.



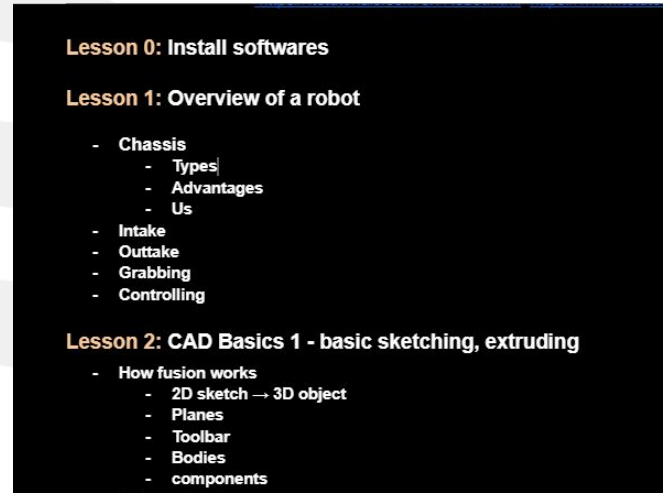
Challenges

Challenges This Year

New Membership

One of our most significant challenges is in retaining new members. We have tried to address the very steep learning curve inherent to robotics by developing lesson plans.

Although we dedicated much of our energy towards the competition this year, we plan to create videos to guide new members through the process, and encourage them to participate next year. Hopefully this will address the gap in “generational knowledge” when our upperclassmen graduate.



The curriculum plans for new members



Jonathan teaches an FLL member about our robot

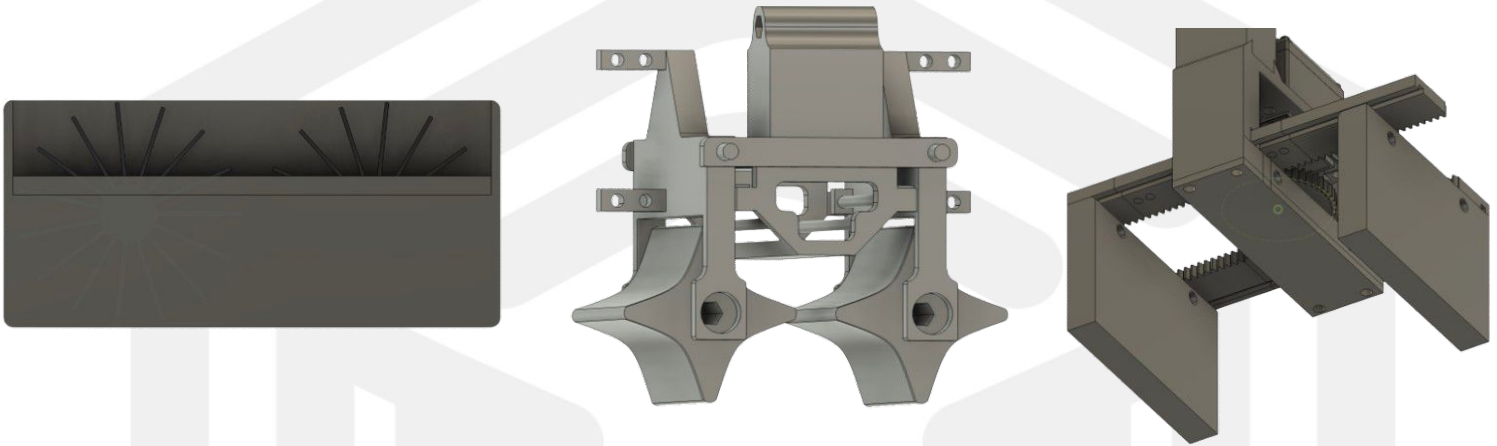
We have also attempted to increase participation in the team by conducting outreach events. These are a combination of events within our school, like bake sales, and events with local organizations, like our MakeInspires event.

By establishing the team in our local community, we hope to encourage more people to follow their passion for STEM, and bring a diverse and collaborative group of people into FIRST, and robotics in general.

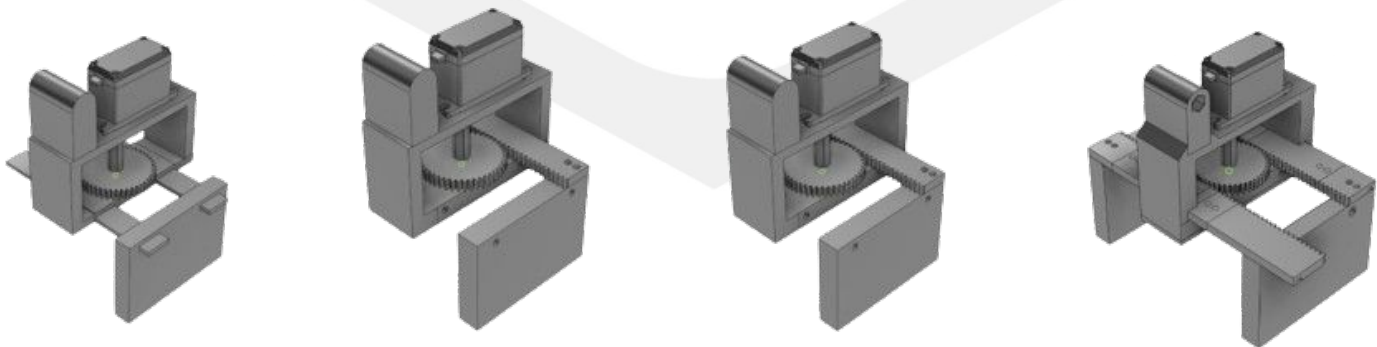
Iterations

Prototypes

These were CAD ideas of intakes before we settled on the claw.



We made many changes to our final claw. Flaws that we fixed in our first prototype was to thicken the steel support bars. The second prototype we chanfered the overlapping pinions and made them a lot thicker. The third prototype we made fixed the bearing to the plate and glued foam to the jaws. The forth and final prototype we set threaded inserts to better secure the claw to the pivot.



Future Plans

Future Team Building

Team 8490 has been participating in the First Tech Challenge since 2014. Because the team has been around for so long, there is generally an established cycle of how knowledge is passed through its members for year to year. The seniors and juniors who are on the team educate new members on the competition and the engineering design process to the younger members of the team. This step is critical to the

team being able to continue operation at its high standard. However, this step was lost due to the COVID-19 pandemic. The effects of this are still being felt, although we have taken extra steps this year to bridge the gap and introduce more people to the world of robotics. Our team created a curriculum, with videos, articles, and other resources to teach incoming members about engineering and coding. This not only encouraged more people to participate who might have been afraid to join, but also helped our team better utilize the skills of its members. We hope to continue this in future seasons.



New members learn the basics of the goBilda system by assembling a second chassis



The team plans to hosts more events like their MakeInspires 2023 robotics class

Next year, we hope to improve our outreach by participating in the STEM fair at the middle school, and improve our engineering design process by referencing this year's engineering notebook. We plan to possibly incorporate unused models that we created this year, particularly the scissor lift, into next year's design. We are still learning as a team, and have made major strides this year to ensure future progress.

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